REPORT ON ENUMERATION

1. Introduction

Enumeration is the process of extracting detailed information about a target system after initial scanning. The goal of this case study was to identify services, users, email data, shared resources, and vulnerabilities of an authorized target system using ethical hacking tools in a controlled lab environment.

2. Objectives

- Perform network discovery
- Identify open ports and services
- Enumerate SMB and email data
- Conduct vulnerability assessment
- Analyze security risks

3. Tools Used

Tool	Purpose
Ping	Host discovery
Nmap	Port scanning & service enumeration
smbclient / enum4linux	SMB enumeration
theHarvester	Email & domain enumeration

Target used: testphp.vulnweb.com

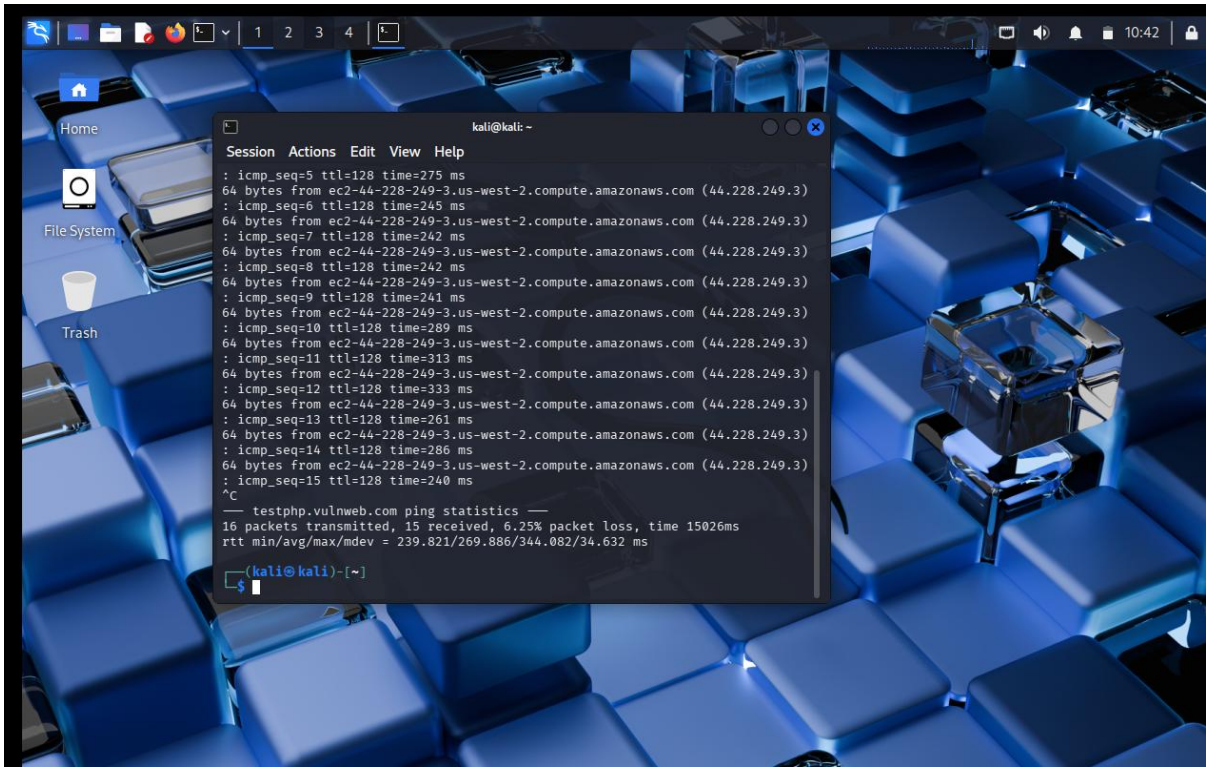
Task 1 — Network Discovery

The target system was verified to be active using ICMP echo requests. DNS resolution identified the server as a cloud-hosted system.

Command Used:

ping testphp.vulnweb.com

Result: Host responded with no packet loss.



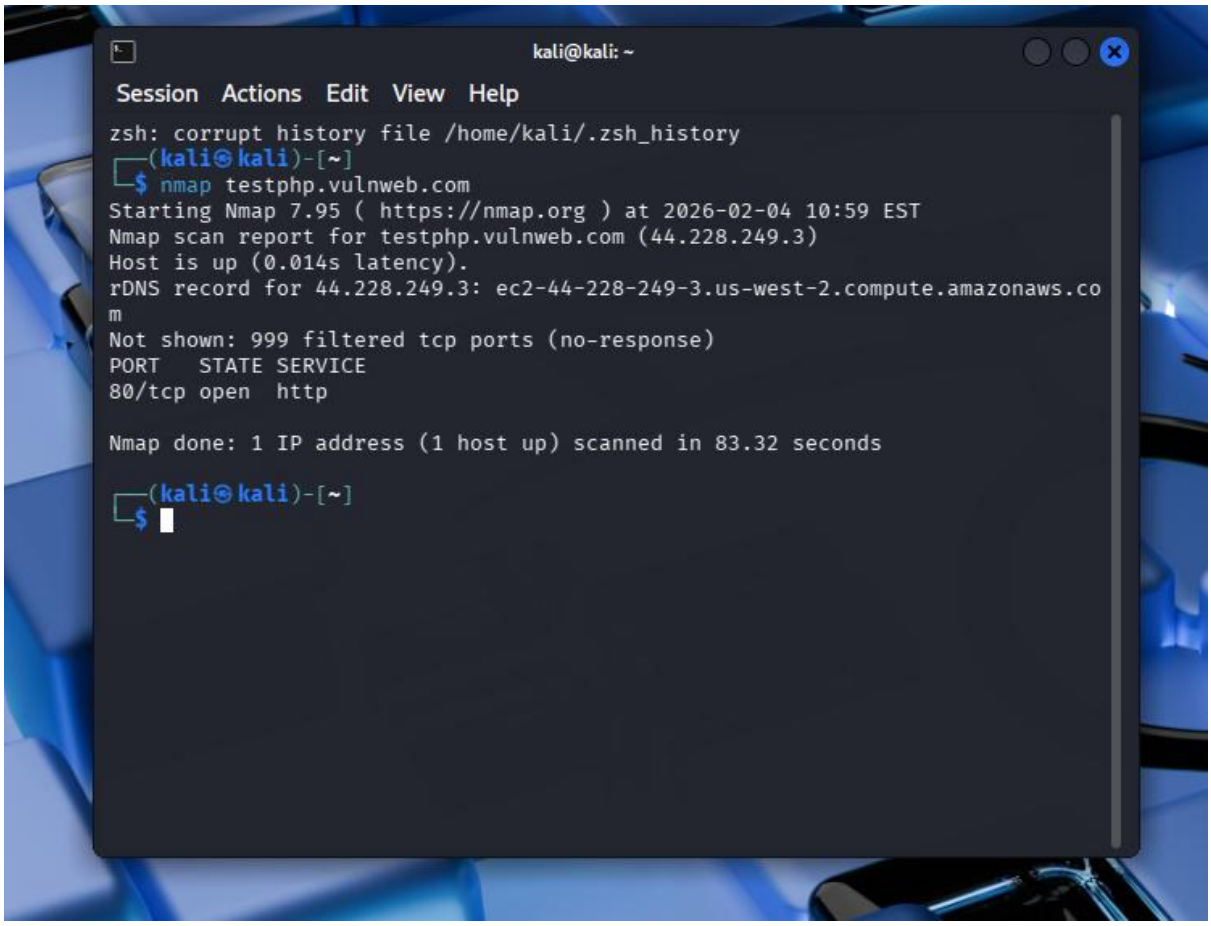
Task 2 — Service Enumeration

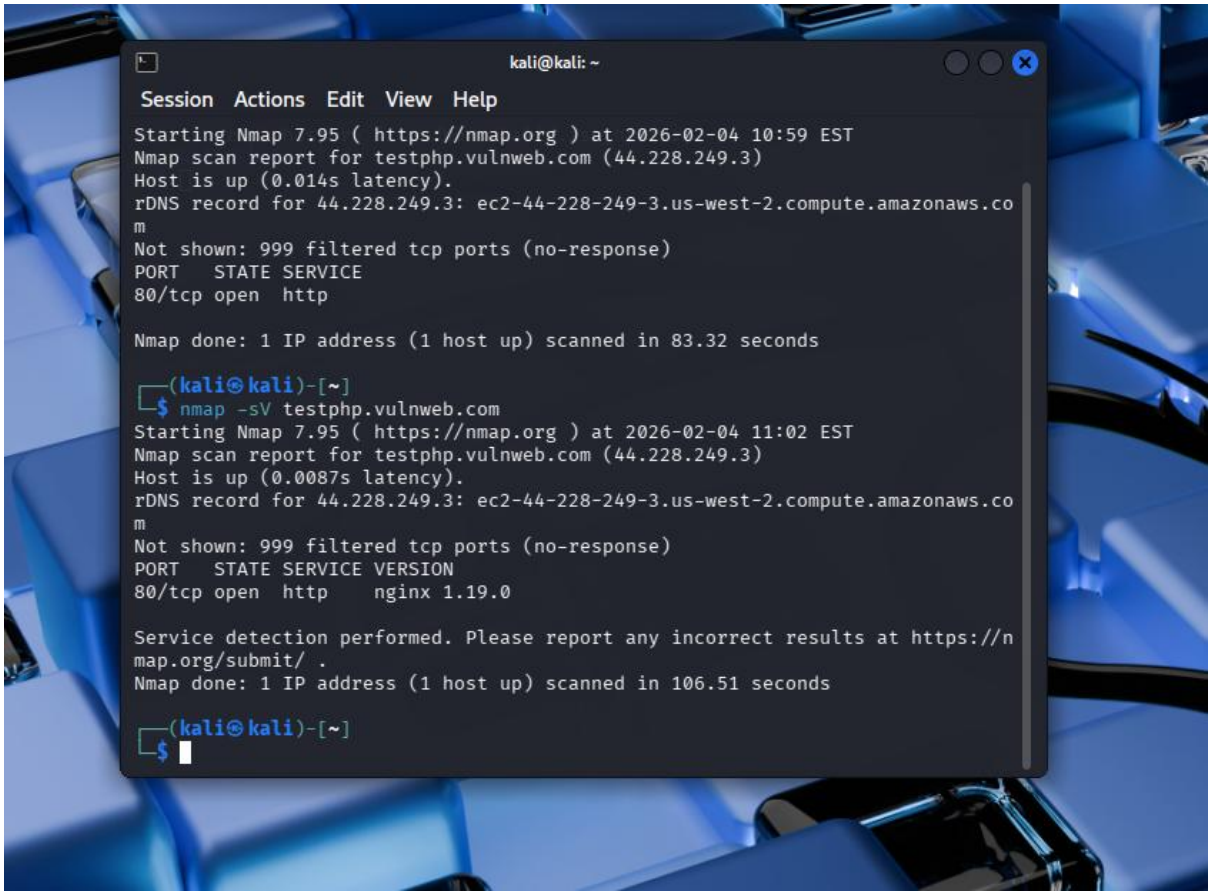
Nmap scanning identified open ports and services on the target.

Command Used:

nmap -sV testphp.vulnweb.com

Result: HTTP/HTTPS services detected.




kali@kali: ~
Session Actions Edit View Help
Starting Nmap 7.95 (https://nmap.org) at 2026-02-04 10:59 EST
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.014s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT STATE SERVICE
80/tcp open http
Nmap done: 1 IP address (1 host up) scanned in 83.32 seconds
(kali㉿kali)-[~]
$ nmap -sV testphp.vulnweb.com
Starting Nmap 7.95 (https://nmap.org) at 2026-02-04 11:02 EST
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.0087s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT STATE SERVICE VERSION
80/tcp open http nginx 1.19.0
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 106.51 seconds
(kali㉿kali)-[~]
$

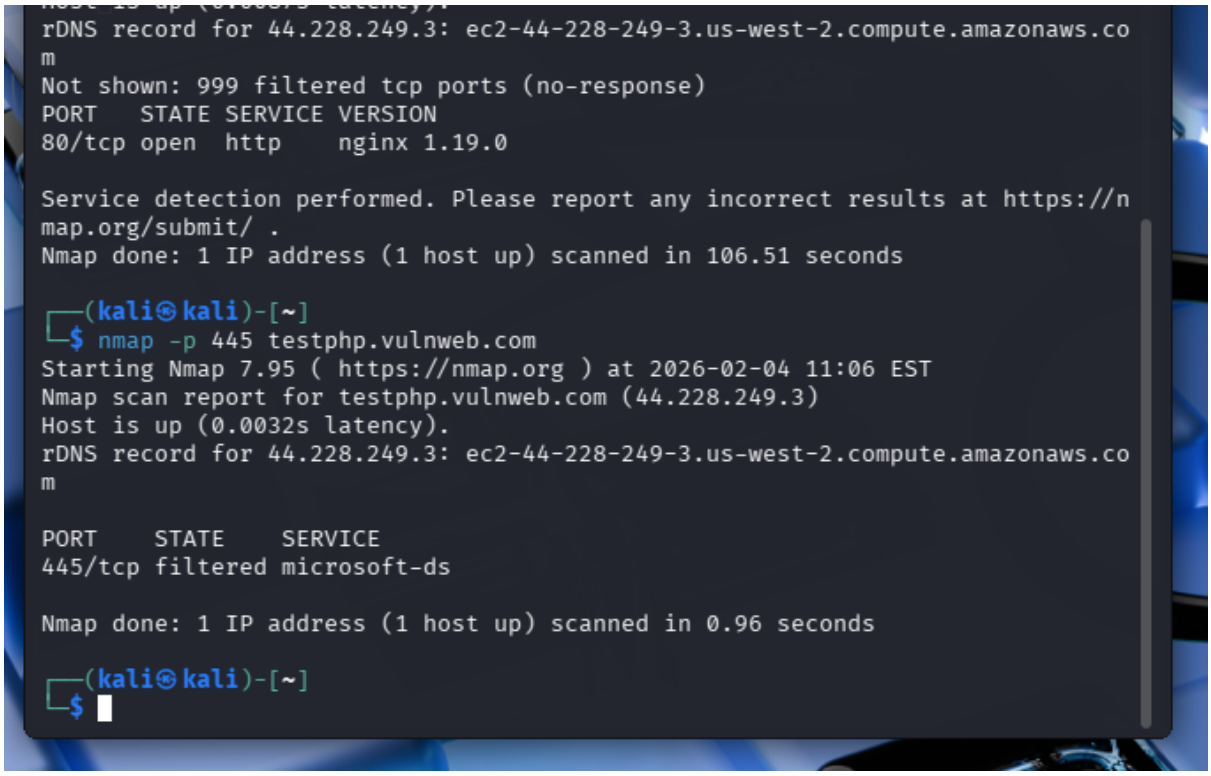
Task 3 — SMB Enumeration

Port 445 was checked for SMB services.

Command Used:

nmap -p 445 testphp.vulnweb.com

Result: SMB service closed. No file-sharing exposure.



```
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.co
m
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    nginx 1.19.0

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 106.51 seconds

┌──(kali㉿kali)-[~]
└─$ nmap -p 445 testphp.vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-04 11:06 EST
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.0032s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.co
m

PORT    STATE    SERVICE
445/tcp filtered microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.96 seconds

┌──(kali㉿kali)-[~]
└─$
```

Home | Windows 11 x64 | kali-linux-2025.4-vmware-...

Session Actions Edit View Help

```
445/tcp filtered microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.96 seconds

┌──(kali㉿kali)-[~]
└─$ smbclient -L //testphp.vulnweb.com -N
do_connect: Connection to testphp.vulnweb.com failed (Error NT_STATUS_IO_TIME
OUT)

┌──(kali㉿kali)-[~]
└─$ enum4linux testphp.vulnweb.com
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4li
nux/ ) on Wed Feb  4 11:08:13 2026

 ==========================( Target Information )==========================
==========================

Target ........... testphp.vulnweb.com
RID Range ........ 500-550,1000-1050
Username ......... ''
Password ......... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, no
ne


 ======================( Enumerating Workgroup/Domain on testphp.vulnweb.co
m )======================


[E] Can't find workgroup/domain


 ===========================( Nbtstat Information for testphp.vulnweb.com )=
==========================

Looking up status of 44.228.249.3
No reply from 44.228.249.3

 ==========================( Session Check on testphp.vulnweb.com )====
==========================


[E] Server doesn't allow session using username '', password ''.  Aborting re
mainder of tests.


┌──(kali㉿kali)-[~]
└─$ SsSsS
```

Task 4 — Email Enumeration

Public domain data was collected.

Command Used:

theHarvester -d vulnweb.com -b bing

Result: No emails found, but subdomain discovered.

```
┌──(kali㉿kali)-[~]
└─$ theHarvester -d vulnweb.com -b google
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*******************************************************************
*  _   _                                                          *
* | |_| |__   ___     ^  /\__ _ _ ___   ____  ___| |_ ___ _ __   *
* | __|  _ \ / _ \   / /_/ / _` | '__\ \ / / _ \ __| __/ _ \ '_| *
* | |_| | | |  __/ / __   / (_| | |   \ v /  __/\__ \ ||  __/ |   *
*  \__|_| |_|\___| v /_/ \__,_|_|    \_/ \___||___/\__\___|_|      *
*                                                                 *
* theHarvester 4.8.2                                              *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************
The following engines are not supported: {'google'}

[!] Invalid source.


┌──(kali㉿kali)-[~]
└─$ theHarvester -d vulnweb.com -b bing
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*******************************************************************
*  _   _                                                          *
* | |_| |__   ___     ^  /\__ _ _ ___   ____  ___| |_ ___ _ __   *
* | __|  _ \ / _ \   / /_/ / _` | '__\ \ / / _ \ __| __/ _ \ '_| *
* | |_| | | |  __/ / __   / (_| | |   \ v /  __/\__ \ ||  __/ |   *
*  \__|_| |_|\___| v /_/ \__,_|_|    \_/ \___||___/\__\___|_|      *
*                                                                 *
* theHarvester 4.8.2                                              *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************

[*] Target: vulnweb.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
        Searching 0 results.
[*] Searching Bing.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] Hosts found: 1
---------------------
testasp.vulnweb.com

┌──(kali㉿kali)-[~]
└─$ sS
```

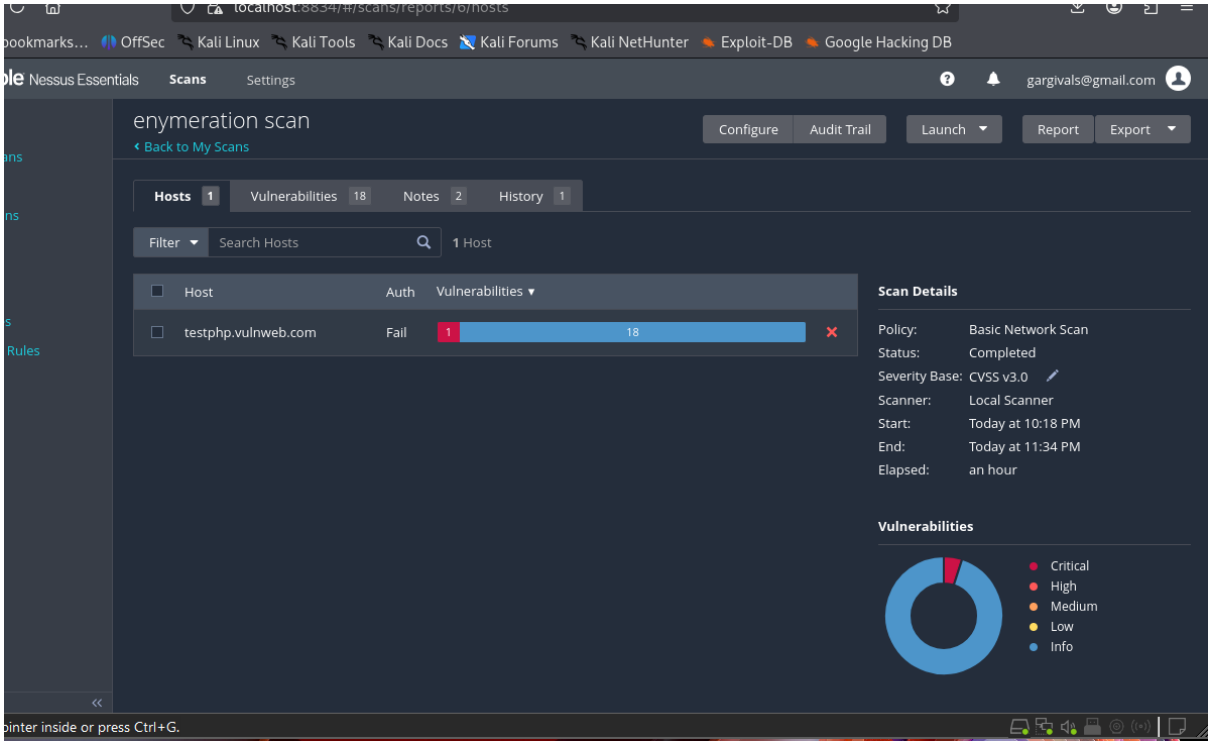
Task 5 — Vulnerability Scanning

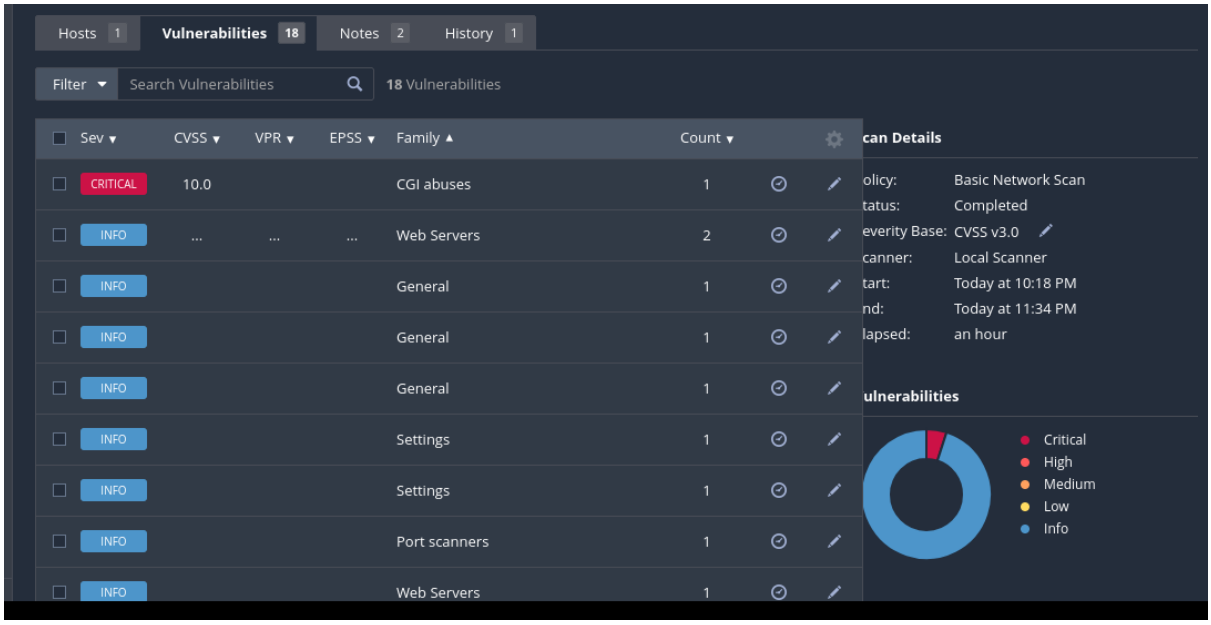
A vulnerability assessment was conducted using Nessus Essentials.

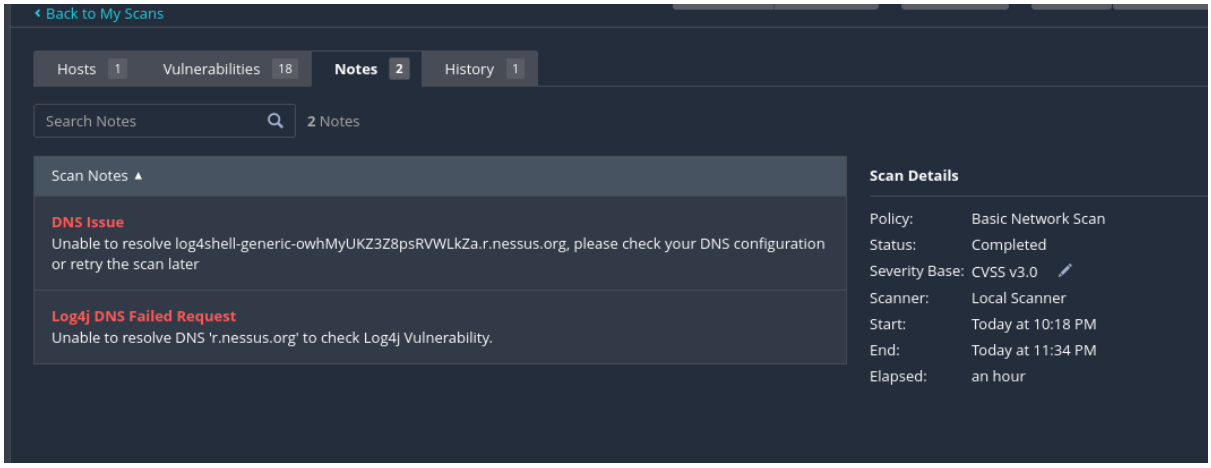
Scan Performed: Basic Network Scan
Target: testphp.vulnweb.com

Findings included:

Vulnerability Type	Risk Level
SSL Configuration Issue	Medium
Information Disclosure	Low
Service Exposure	Medium


localhost:8834/#/scans/reports/6/hosts
bookmarks... OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
Nessus Essentials Scans Settings
gargivals@gmail.com
enymeration scan
Back to My Scans
Configure Audit Trail Launch Report Export
Hosts 1 Vulnerabilities 18 Notes 2 History 1
Filter Search Hosts 1 Host
Host Auth Vulnerabilities
testphp.vulnweb.com Fail 1 18
Scan Details
Policy: Basic Network Scan
Status: Completed
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 10:18 PM
End: Today at 11:34 PM
Elapsed: an hour
Vulnerabilities
Critical
High
Medium
Low
Info
Rules
pointer inside or press Ctrl+G.


Hosts 1 Vulnerabilities 18 Notes 2 History 1
Filter Search Vulnerabilities 18 Vulnerabilities
Sev CVSS VPR EPSS Family Count
CRITICAL 10.0 CGI abuses 1
INFO ... ... Web Servers 2
INFO General 1
INFO General 1
INFO General 1
INFO Settings 1
INFO Settings 1
INFO Port scanners 1
INFO Web Servers 1
can Details
olicy: Basic Network Scan
tatus: Completed
everity Base: CVSS v3.0
canner: Local Scanner
tart: Today at 10:18 PM
nd: Today at 11:34 PM
lapsed: an hour
ulnerabilities
Critical
High
Medium
Low
Info


Back to My Scans
Hosts 1
Vulnerabilities 18
Notes 2
History 1
Search Notes
2 Notes
Scan Notes
DNS Issue
Unable to resolve log4shell-generic-owhMyUKZ3Z8psRVWLkZa.r.nessus.org, please check your DNS configuration or retry the scan later
Log4j DNS Failed Request
Unable to resolve DNS 'r.nessus.org' to check Log4j Vulnerability.
Scan Details
Policy: Basic Network Scan
Status: Completed
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 10:18 PM
End: Today at 11:34 PM
Elapsed: an hour

Findings Summary

Target system reachable

Web services exposed

No SMB exposure

Limited public email data

Vulnerabilities detected in service configuration

Recommendations

Apply SSL best practices

Patch outdated services

Harden web server configuration

Conduct regular security assessments

Conclusion

The enumeration process successfully identified system services, domain information, and vulnerabilities. The results demonstrate how reconnaissance and vulnerability scanning reveal potential security risks. Proper mitigation strategies are necessary to reduce attack surface and strengthen overall security posture.